

Algorithms for genomic data analysis

Introduction to next generation sequencing and its applications

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Organizational remarks

- Joint course for bioinformaticians and computer scientists
- Teachers:
 - Norbert Dojer
 - Aleksander Jankowski
- Final assessment:
 - assignments
 - may be carried out in teams ≤ 3 persons
 - final score based on 2 out of 3 assignments
 - (optionally) oral exam

Materials

Course on Moodle:

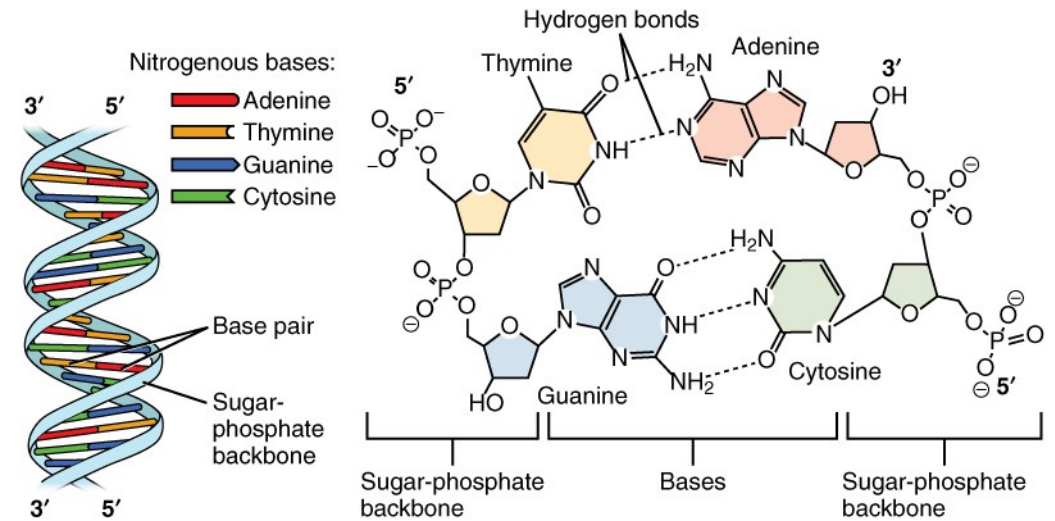
<https://moodle.mimuw.edu.pl/course/view.php?id=2255>

registration code: **cotR462F**

Teaching materials by Ben Langmead:

<https://www.langmead-lab.org/teaching.html>

This diagram illustrates the flow of genetic information. At the top, a yellow sphere represents the **Cell**, containing a blue nucleus with yellow chromosomes. A bracket labeled **Genome** encompasses the chromosomes. Below, a yellow chromosome is shown unraveling into a double helix structure labeled **DNA**. A bracket labeled **Genes** identifies specific segments of the DNA. The DNA helix is color-coded with nitrogenous bases: Adenine (A), Thymine (T), Guanine (G), and Cytosine (C). At the bottom, a large grey arrow points from the DNA to a series of 3D protein structures. One structure is labeled **Protein**. The proteins are shown in various colors (green, red, blue) and shapes, representing different functional molecules.



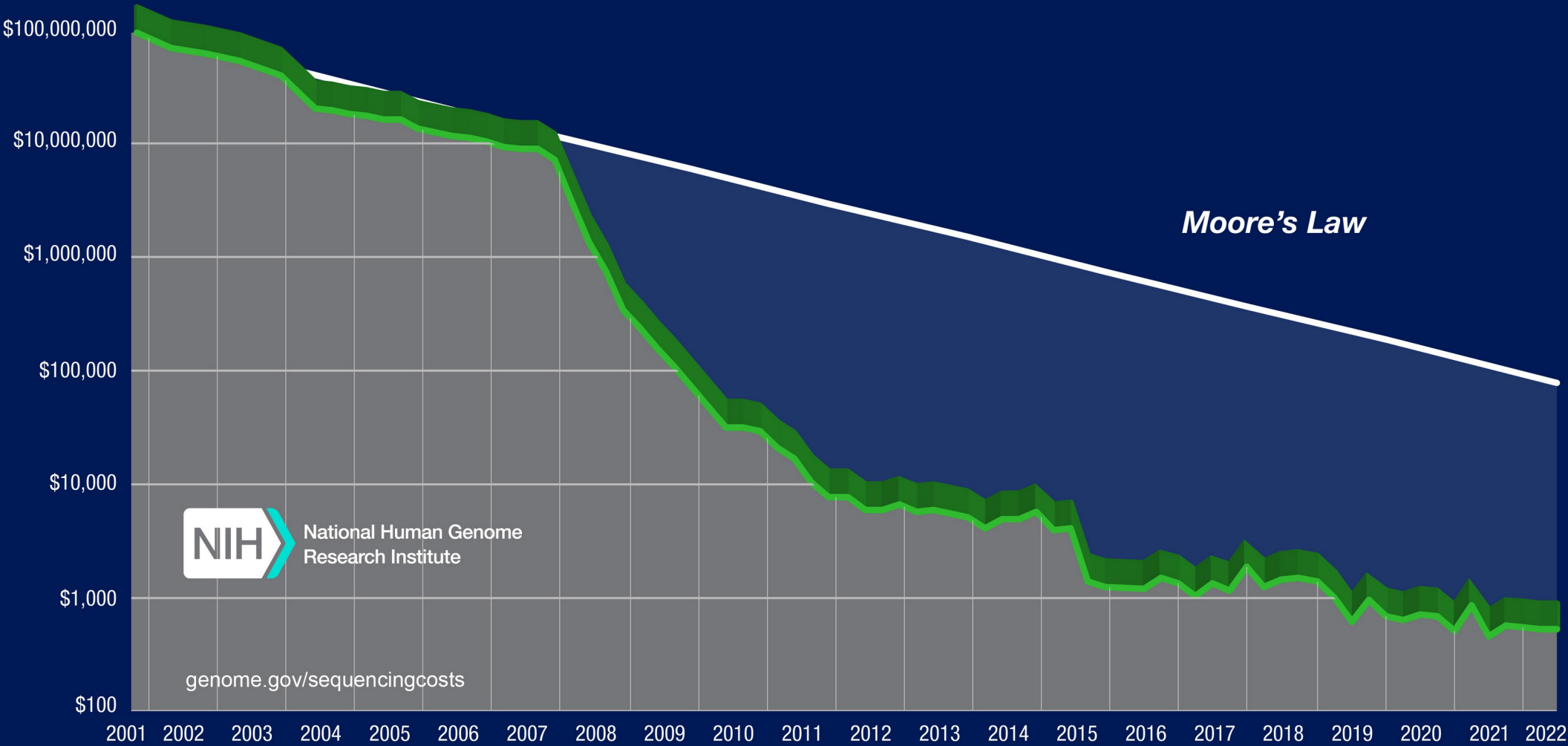
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DNA sequencing

DNA sequencing is the process of determining the precise order (sequence) of nucleotides within a DNA molecule.

- Before sequencing, DNA molecules are fragmented, selected, amplified etc.
- Sequencing output consists of reads of typical length 50 – 20,000 bp, corresponding to whole fragments or their ends
- Various sample preprocessing protocols yield data that is suitable for various applications and require various analysis methods

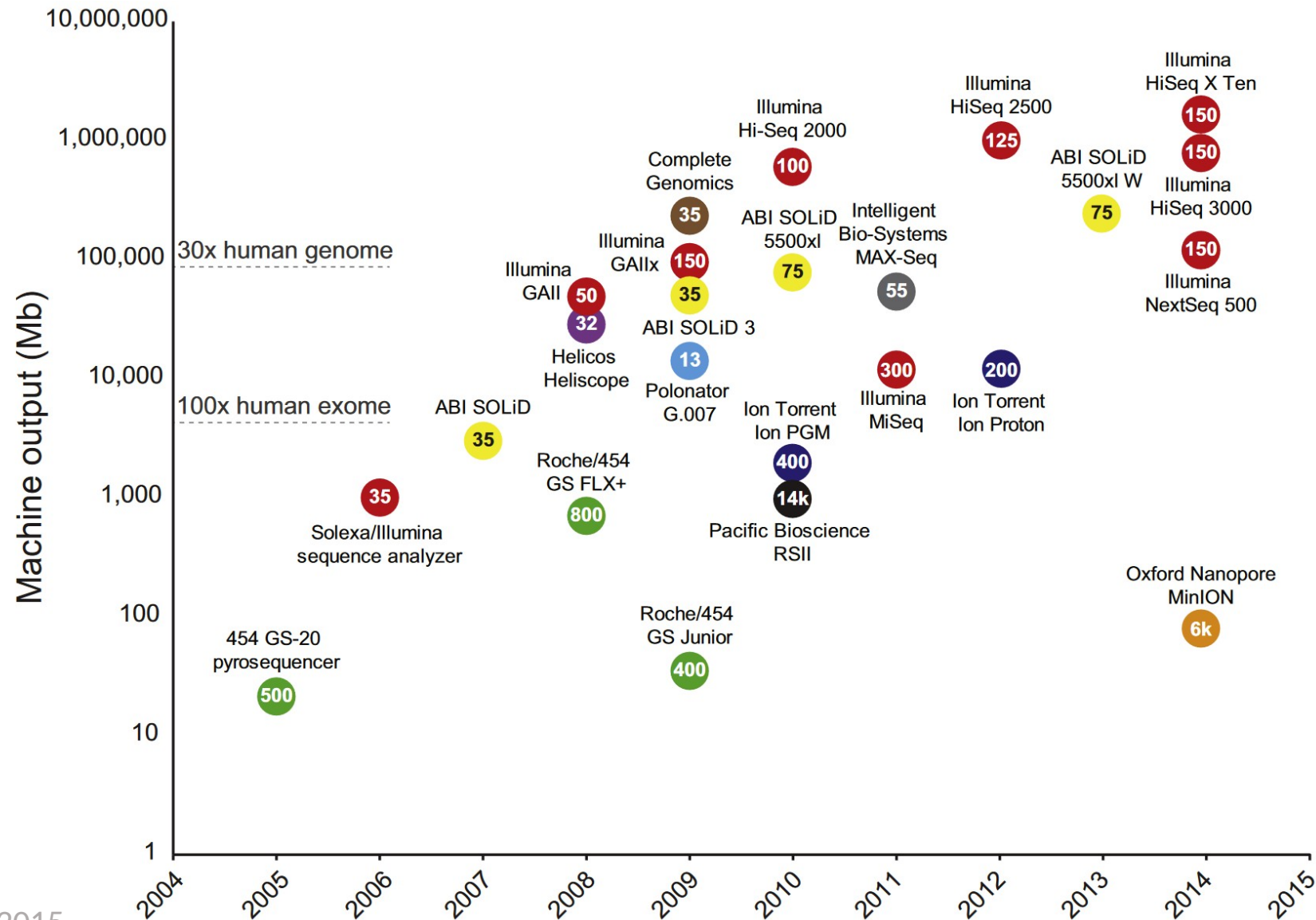
Cost per Human Genome



Technology generations

- First generation sequencing: Sanger sequencing
 - very expensive
- Next (second) generation sequencing: 454, SOLiD, Illumina, IonTorrent
 - high-throughput
- Third generation sequencing: Pacific Bioscience, Oxford Nanopore
 - very long reads
 - single molecule sequencing

Sequencing technologies



Technology comparison

Sanger technology

- read length: 700–1000bp
- accuracy: >99.9%
- runtime: 2h
- throughput: ≤ 96 reads per run
- cost: 500,000\$/Gb

Current

Platform	Read Length (bp)	Accuracy (%)	Run Time	Bases Per Run (Gb)	Cost/Gb
454 Roche	1000	99	24 h	0.54	\$10,000
SOLiD	75	99.9	7 d	520	\$10
Illumina	300	99.9	3 d	1800	\$10
Ion Torrent	400	99	2 h	15	\$100
Pacific Bioscience	20,000	90	3 h	12,000	\$600
Oxford Nanopore	10,000	90	2 d	42	\$1000

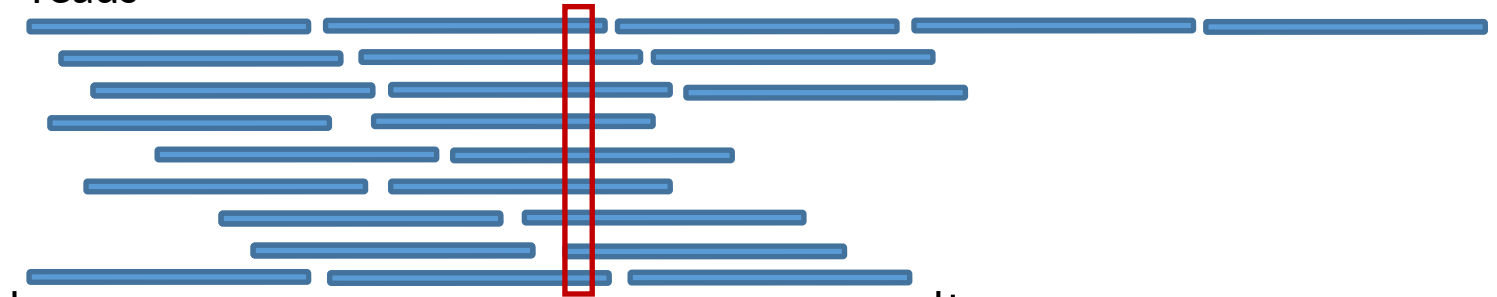
bp, base pairs; Gb, gigabase pairs; h, hours; d, days.

Key concepts: coverage

multiple copies of a genome



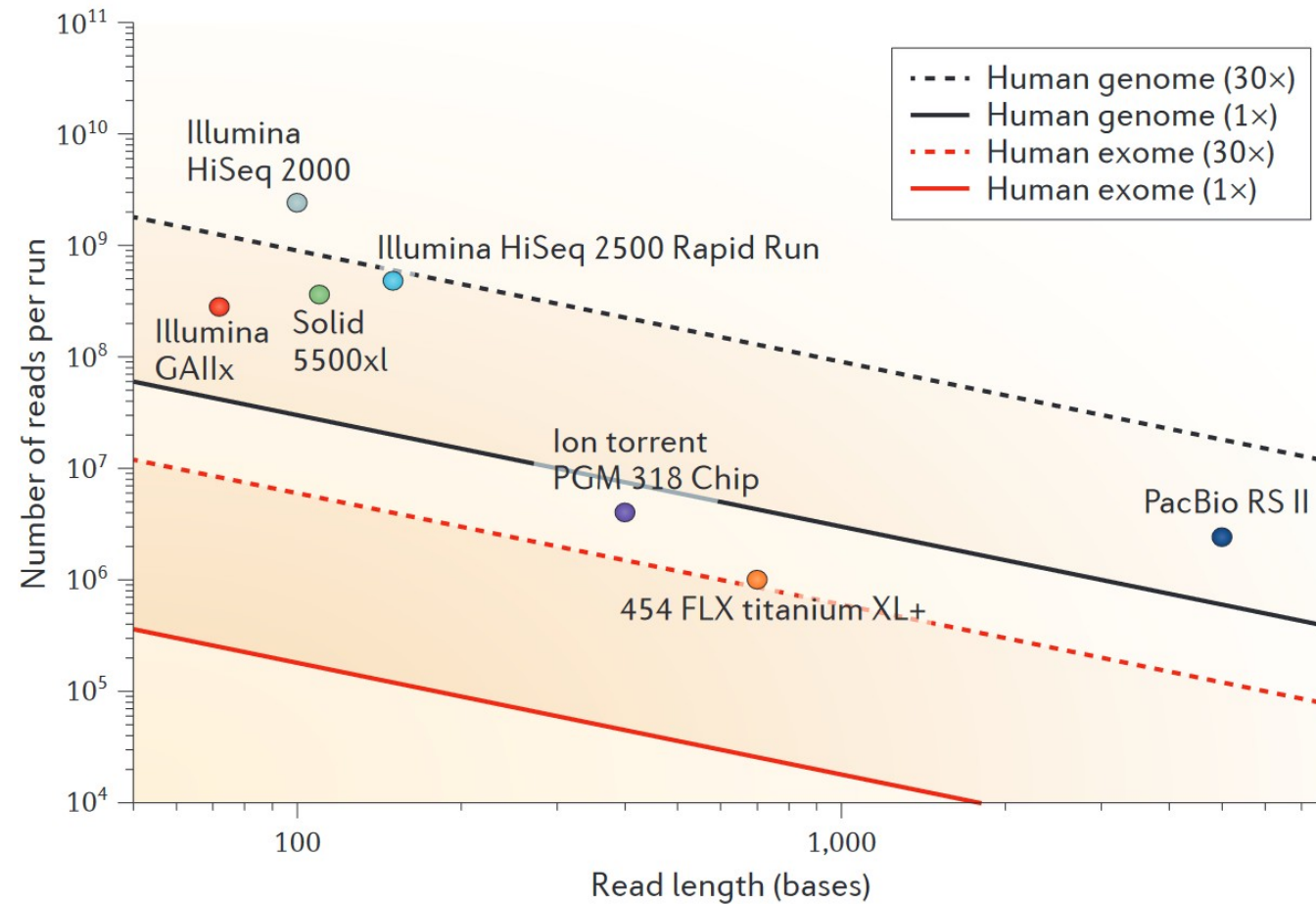
reads



high coverage

low coverage

Key concepts: coverage



GAllx, Genome Analyzer IIX; PacBio, Pacific Biosciences; PGM, personal genome machine.

Key concepts: single- and paired-end reads

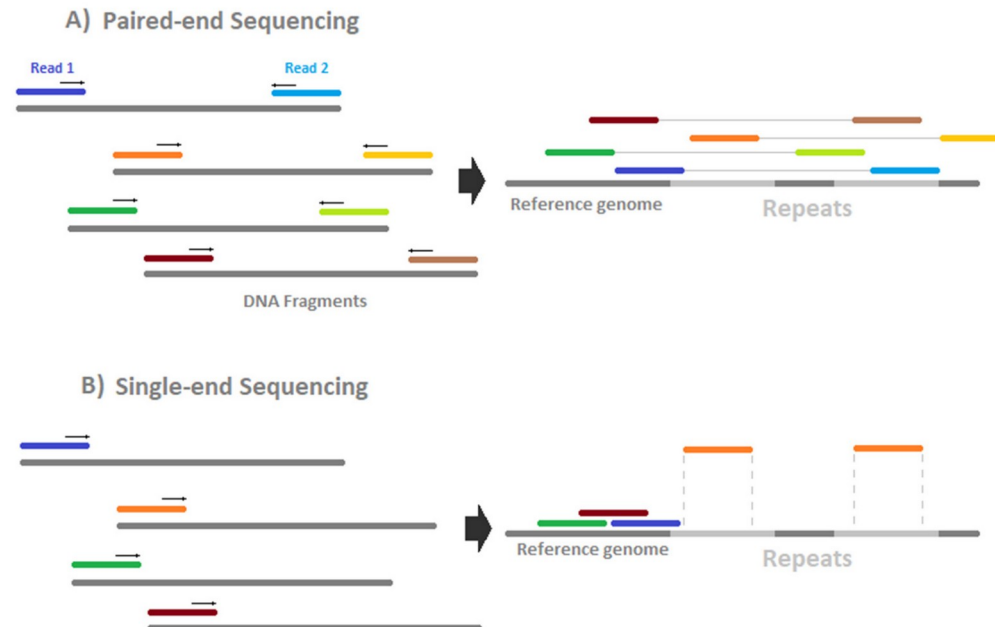
Single-read sequencing

Sequencing DNA from only one end: simple, economical.

Paired-read sequencing

Sequencing both ends of a fragment: higher quality data.

Facilitates detection of genomic rearrangements, repetitive sequence elements, gene fusions and novel transcripts.



Applications of DNA sequencing

